

Development of Autonomous Navigation System for Mobile Robots in Indoor Environment Using Navigation Beacon

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PROBLEM DEFENITION

Design an optimal arrangement and minimal deployment of ArUco markers to enable robust and reliable indoor robot navigation while reducing infrastructure requirements.

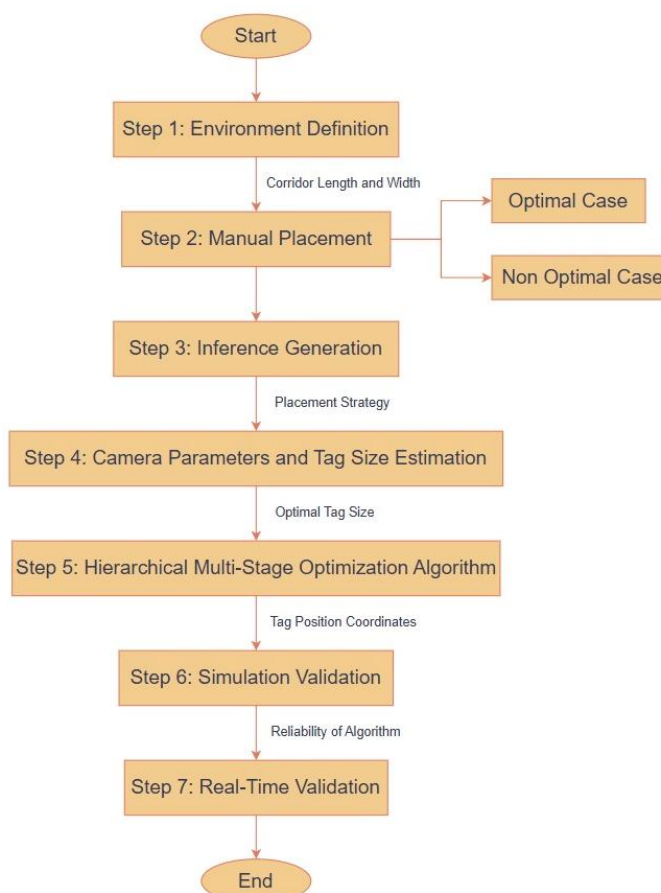
Impact on Sustainable Development Goals (SDGs)

- **SDG 9:** Low-cost, GPS-free indoor navigation for industrial automation.
- **SDG 11:** Enables efficient autonomous robots in public spaces (e.g., hospitals, airports).
- **SDG 12:** Reduces energy use and hardware through optimized tag placement.

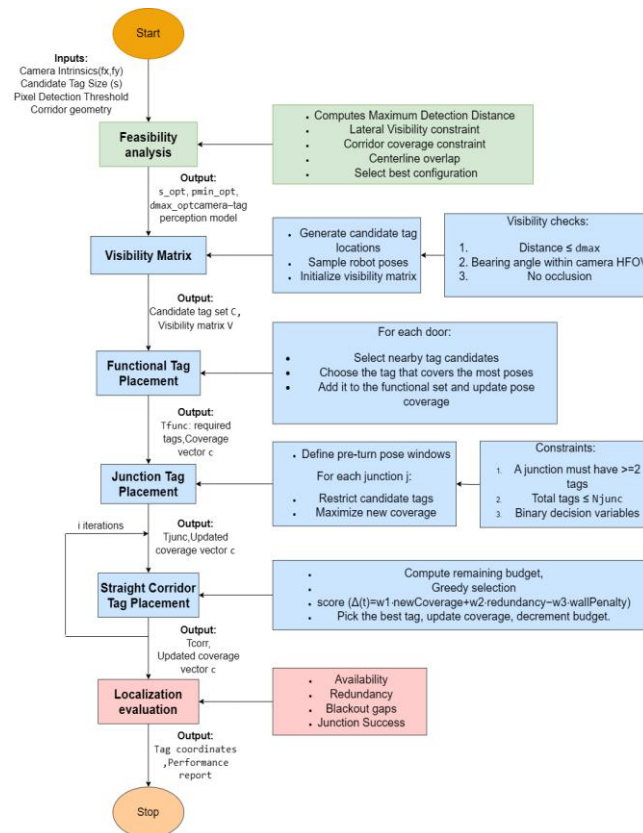
OBJECTIVES / DELIVERABLES

- Calibrate the camera for accurate marker-based localization
- Develop an optimized ArUco marker placement strategy
- Design a multi-stage placement approach (feasibility, visibility, greedy optimization)
- Validate performance in simulation and real-world corridor

METHODOLOGY

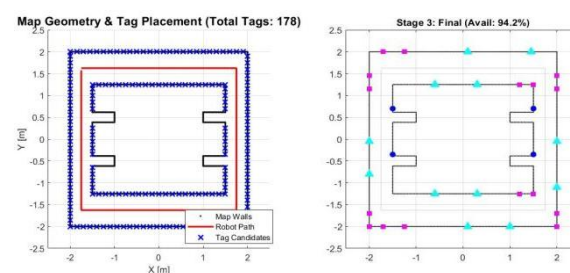


Flowchart



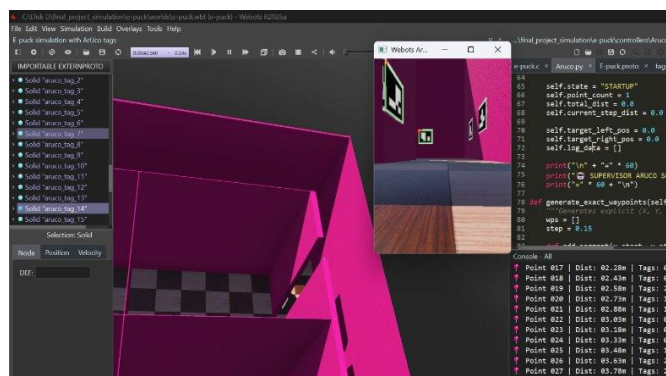
Proposed Algorithm

RESULTS AND DISCUSSION



Available tags & Chosen tags

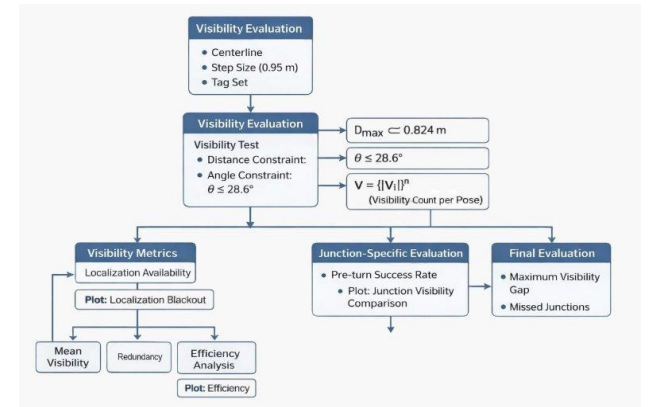
- 94.2% localization availability
- 82% reduction in number of tags
- 100% pre-turn success rate
- Mean visibility - 1.01 tags/frame



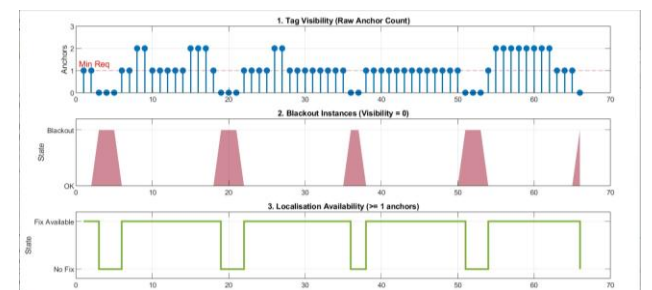
Webots Simulation



Jetson Nano bot



Flowchart of Evaluation



Evaluation plots of the result table

Key Insight: Visibility-aware placement achieves higher localization with fewer tags compared to uniform deployment.

A visibility-aware, multi-stage

CONCLUSION

Optimization approach for ArUco marker placement is proposed, significantly improving indoor robot navigation. The method achieves high localization reliability (94.2%) while reducing infrastructure by 82%, and demonstrates consistent performance across simulation and real-world validation. This approach provides a scalable and cost-effective solution for deploying autonomous robots in structured indoor environments.

KEY REFERENCES

- Garrido-Jurado, S., Muñoz-Salinas, R., Madrid-Cuevas, F. J., & Marín-Jiménez, M. J. (2014). Automatic generation and detection of highly reliable fiducial markers under occlusion. Pattern Recognition.
- Romero-Ramirez, F. J., Muñoz-Salinas, R., & Medina-Carnicer, R. (2018). Speeded up detection of squared fiducial markers. Image and Vision Computing.
- Krajník, T., Vonásek, V., Fiser, D., & Faigl, J. (2021). ARTag-based localization for mobile robots in indoor environments. Robotics and Autonomous Systems.